

## 8-LAYER ENTERPRISE ARCHITECTURE

### LAYER 1 — DATA SOURCES (12 SYSTEMS)

**Internal** Guidewire Claims · Duck Creek Policy · TriZetto Medical Billing · SAP Finance · Salesforce CRM · Workday HR/Benefits · Sedgwick WC TPA  
**External** ISO ClaimSearch · LexisNexis Risk · NICB · CMS Provider Exclusions · NOAA Weather · DMV Feeds · OFAC Sanctions · Court Records · IRS TIN

### LAYER 2 — INGESTION PIPELINE (SIMULATED KAFKA)

**Topics** claims.fnol · claims.update · policy.change · medical.bill · repair.invoice · payment.issued · iso.claimsearch · external.enrichment  
**Pattern** Batch (nightly 100K) + Real-time (<500ms per claim) · Same MERGE logic · Swap real Kafka in Phase 6

### LAYER 3 — MULTI-MODAL AI PROCESSING

**CV Pipeline** Damage photos · Repair invoices · Accident scenes · Medical docs  
**5 CV Layers** EXIF/Metadata · ELA+FFT Pixel Forensics · Duplicate Detection · Cross-Document Consistency · Temporal Consistency  
**OCR+LLM** Entity extraction → graph loading · Anomaly flagging · Template similarity detection  
**Synthetic ID** Discriminative detection first → generative detection capability designed in

### LAYER 4 — ENTITY RESOLUTION

**Cross-system** Master person\_id across Guidewire + Salesforce + Workday + Sedgwick  
**Fuzzy match** Names · Phones · Addresses · DOB · SSN-last4 · Device fingerprint  
**ISO match** Cross-carrier identity resolution via ISO ClaimSearch xref

### LAYER 5 — GRAPH STORE (NEO4J AURADB)

**24 node labels** Claim · ClaimVersion · CoverageLine · ClaimPayment · Person · Employee · Dependent · Policy · Vehicle · Fleet · MedicalProvider · MedicalBill · RepairShop · Attorney · TowCompany · Employer · Address · Phone · CrossSystemID · Adjuster · BankAccount · LitigationEvent · InjuryRecord · FraudRing  
**Temporal** Properties + timestamps on all nodes · GDS temporal algorithms · Time-series graph for complex patterns (Phase 2)

### LAYER 6 — INTELLIGENCE ENGINE

**GDS** Connected Components · Louvain Community · PageRank · Betweenness Centrality · Temporal WCC  
**GNN** PyTorch Geometric designed in · Optional activation · Heterogeneous graph neural network for collusion  
**Scoring** 21 fraud patterns · LOB-specific weights · Payment hold logic · Cross-carrier signals from ISO

## LAYER 7 — AGENTIC AI (HUMAN-IN-THE-LOOP)

**Co-pilot** AI assembles evidence package · Suggests investigation steps · Drafts SIU referral narrative

**Human** Analyst reviews · Approves/rejects referral · Adds external evidence · Closes case

**Autonomous** Low-risk only: duplicate detection · OFAC screening · Payment hold triggers

## LAYER 8 — INVESTIGATOR DASHBOARD

**Views** Graph ring visualization · Temporal timeline · Geospatial map · Adjuster workqueue · Cross-carrier hits

**Actions** SIU referral · Payment hold · Watchlist flag · Case notes · Evidence upload

## 21 FRAUD PATTERNS ACROSS 6 CATEGORIES

### Cat 1 — Healthcare/Medical

**HIGH** Medical mill (1 doctor → 50+ claims)

**HIGH** CPT upcoding (billed vs allowed ratio)

**HIGH** Phantom treatment (no facility record)

**MED** Duplicate billing across carriers

**MED** Unbundling CPT codes

**LOW** Exaggerated soft tissue injury

### Cat 3 — Temporal/Sequence

**HIGH** Claim velocity (3+ claims in 90 days)

**HIGH** Impossible timeline (treatment before incident)

**MED** Reserve pattern anomaly (rapid escalation)

**MED** Policy lapse/reinstate cycle before claim

**LOW** Seasonal fraud clustering

### Cat 5 — Insider/Adjuster

**HIGH** Adjuster + shop collusion (approval rate anomaly)

**HIGH** Adjuster + attorney pipeline (settlement inflation)

**MED** Phantom payment (payee = adjuster network)

**MED** Reopened closed claims pattern

### Cat 2 — Property-Specific

**HIGH** Arson ring (multi-property, same owner/agent)

**HIGH** Flood fraud (pre-existing damage claimed)

**MED** Contractor collusion (inflated estimates)

**MED** Contents inflation (no purchase records)

**LOW** Post-loss policy inception

### Cat 4 — Identity & Synthetic

**HIGH** Synthetic identity (thin credit file + new SSN)

**HIGH** Identity takeover (address/phone sudden change)

**MED** Phantom passenger recycling

**LOW** Address fraud (rate territory mismatch)

### Cat 6 — Cross-Carrier (ISO)

**HIGH** Same incident filed at 2+ carriers

**HIGH** Serial claimant across carriers (ISO hit)

**MED** Provider excluded at one carrier still billing another

**MED** Cross-carrier medical mill network

**BUILD PHASES — PRODUCTION ROADMAP**

| PHASE                   | DELIVERABLE                   | KEY COMPONENTS                                                                        | AURADB NODES          |
|-------------------------|-------------------------------|---------------------------------------------------------------------------------------|-----------------------|
| A — Schema              | Production Neo4j schema       | 24 node labels · 32 relationships · all constraints · temporal indexes                | 0 (schema only)       |
| B — Data Generator      | 50K node production dataset   | 5 source systems · 21 fraud patterns · 10 years · Hartford geography · ISO simulation | ~50K nodes            |
| C — Entity Resolution   | Cross-system ID pipeline      | Fuzzy match · master ID · CrossSystemID nodes · ISO xref                              | +5K xref nodes        |
| D — Intelligence Engine | GDS + scoring all 21 patterns | Temporal GDS · LOB weights · payment hold · ISO signal · GNN design-in                | +score properties     |
| E — Multi-Modal AI      | CV + OCR + LLM pipeline       | 5-layer CV · entity extraction · anomaly detection · synthetic ID discriminator       | +image metadata nodes |
| F — Agentic Co-pilot    | Human-in-loop investigator    | Evidence assembly · SIU narrative · analyst approval workflow                         | +case nodes           |
| G — Dashboard v2        | Production investigator UI    | Timeline · geo map · cross-carrier view · adjuster queue · evidence panel             | read only             |
| H — GNN (Optional)      | PyTorch Geometric collusion   | Heterogeneous GNN · trained on labeled rings · replaces/augments GDS Louvain          | read only             |